

Drones with Artificial Intelligence

Prof. Dr. Christian Baun – christianbaun@fra-uas.de

Further information:

www.christianbaun.de/Master_Projekt_WS2627/

Participants in the project work in groups of up to four students. Each group is provided with a fully assembled and flight-ready FPV drone as well as the required hardware and software components. **The objective of the project is to independently develop, implement, and experimentally evaluate a drone-based solution for the autonomous exploration and mapping of indoor environments.** The drone should explore rooms and corridors of a building as autonomously as possible and create at least a two-dimensional map of its surroundings. A three-dimensional representation may be implemented as an extension. Since reliable GPS reception is generally unavailable inside buildings, alternative methods for positioning, navigation, and obstacle detection must be investigated and combined.

The project comprises the following tasks:

1. **Drone Platform and System Architecture.** Become familiar with the provided drone platform, flight controller, radio link, video transmission system, and available interfaces. Design a suitable overall architecture and decide which functions are to be executed on the drone and which on an external ground station.
2. **Autonomous Indoor Flight.** Preferably use ArduPilot to enable stable and safe flight without GPS. Combine suitable sensors such as LiDAR rangefinders, optical-flow sensors, cameras, and inertial sensors. Manual takeover or a safe shutdown must remain possible at all times.
3. **Localization and Mapping.** Implement a method for simultaneous localization and mapping. The minimum requirement is a two-dimensional occupancy map or floor plan. A three-dimensional point cloud or semantic map may be added as an extension. Suitable technologies include ROS 2, OpenCV, RTAB-Map, ORB-SLAM3, and Cartographer.
4. **Autonomous Exploration and Navigation.** The drone should identify unexplored areas, select suitable exploration targets, and plan collision-free routes. Rooms, corridors, doors, dead ends, and obstacles should be taken into account. After completing the exploration or in the event of an error, the drone should land safely or return to a known starting point. Optionally, objects may be detected and marked on the map.
5. **Integration and Mechanical Modifications.** Integrate the required cameras, range sensors, and, if necessary, a single-board computer into the drone. Design and manufacture suitable mounts or frame extensions. A 3D printer is available for producing custom components.
6. **Experimental Evaluation.** Evaluate the complete system using reproducible experiments. Suitable metrics include map quality, localization error, area coverage, exploration time, the number of manual interventions, computing and energy requirements, and robustness under difficult conditions such as changing lighting, reflective surfaces, textureless walls, narrow passages, and moving obstacles.
7. **Documentation and Presentation.** Create complete online documentation that enables other students, researchers, and lecturers to reproduce and further develop the project. Instead of a traditional PDF report or slide presentation, document the architecture, implementation, experiments, results, and limitations online, for example using GitHub Pages. Present the results with a poster and a live demonstration.

The focus is not solely on a successful demonstration, but equally on a systematic approach, reproducible experiments, and a well-founded analysis of the technical limitations, particularly regarding computing power, battery capacity, drift, incomplete sensor data, and communication failures.